

Yolov8 模型转换与部署



版本历史

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1. 环境搭建

1.1. 安装 python 环境

(1) 安装 Miniconda

conda 允许用户设置独立的环境来运行不同版本的 Python 或其他语言, 而无需切换到其他环境管理器。Miniconda 和 Anaconda 均是 conda 工具, 但 Miniconda 是 Anaconda 最小安装程序, 内存占用较小, YOLOv8 的训练环境搭建将使用 conda 工具创建一个独立的 python 环境。

执行下面命令安装 Miniconda3

```
sudo apt-get update

wget

https://mirrors.tuna.tsinghua.edu.cn/anaconda/miniconda/Miniconda3-latest-Linux-x86_64

.sh

chmod +x Miniconda3-latest-Linux-x86_64.sh

./Miniconda3-latest-Linux-x86_64.sh -b -p ~/miniconda3
```

```
sunshine@sunshine-vm:~/Workspace/Ascend$ wget https://mirrors.tuna.tsinghua.edu.cn/anaconda/miniconda/Miniconda3-latest-Linux-x86_64.sh
--2025-09-02 09:49:29-- https://mirrors.tuna.tsinghua.edu.cn/anaconda/miniconda/Miniconda3-latest-Linux-x86_64.sh
正在连接 192.168.101.2:7890... 已连接。
已发出 Proxy 请求, 正在等待回应... 200 OK
长度: 162129736 (155M) [application/octet-stream]
正在保存至: "Miniconda3-latest-Linux-x86_64.sh"
Miniconda3-latest-Linux-x86_64.sh 100%[=====] 154.62M 11.3MB/s 用时 16s
2025-09-02 09:49:44 (9.95 MB/s) - 已保存 "Miniconda3-latest-Linux-x86_64.sh" [162129736/162129736]
sunshine@sunshine-vm:~/Workspace/Ascend$

sunshine@sunshine-vm:~/Workspace/Ascend$ ./Miniconda3-latest-Linux-x86_64.sh -b -p ~/miniconda3
PREFIX=/home/sunshine/miniconda3
Unpacking bootstrapper...
Unpacking payload...

Installing base environment...

Preparing transaction: ...working... done
Executing transaction: ...working... done
installation finished.
sunshine@sunshine-vm:~/Workspace/Ascend$
```

(2) conda 命令创建虚拟环境

下面命令为 conda 环境的基本命令。

```
#初始化 miniconda

cd ~/miniconda3/bin/

./conda init

./conda config --set auto_activate false

#重载环境

source ~/.bashrc

#列出所有环境

conda env list

#创建虚拟环境,[env_name]为环境名，这里选择 python 版本为 3.10，可选择其他版本，
#但需满足 python>=3.8

conda create -n [env_name] python=3.10

#切换环境

conda activate [env_name]

#退出环境

conda deactivate

#删除环境

conda env remove -n [env_name]
```

2. 模型转换

2.1. 安装 yolov8 的 python 环境

(1) conda 创建 yolov8 环境

```
conda create -n yolov8 python=3.10

conda activate yolov8
```

(2) 安装 yolov8 相关依赖

创建 requirements_yolov8.txt 文件，添加以下内容。

```
# Usage: pip install -r requirements_yolov8.txt

# Base ----- matplotlib>=3.2.2
numpy==1.26.4
opencv-python==4.8.1.78
Pillow>=7.1.2
PyYAML>=5.3.1
requests>=2.23.0
scipy>=1.4.1
torch==2.1.0
torchvision==0.16.0
tqdm>=4.64.0

# Plotting ----- pandas>=1.1.4
seaborn>=0.11.0

# Export ----- onnx>=1.12.0 # ONNX export
onnxsim>=0.4.1 # ONNX simplifier

# Extras ----- psutil # system utilization
thop>=0.1.1 # FLOPs computation

#-----
```

在 conda 的 yolov8 环境下安装。命令后加-i 指定源。

```
pip install --upgrade pip
pip install -r requirements_yolov8.txt
pip install ultralytics
```

2.2. 安装 atc 的 python 环境

(1) conda 创建 atc 环境

```
conda create -n atc python=3.9.2
conda activate atc
```

(2) 安装 atc 相关依赖

在 conda 的 atc 环境下安装。命令后加-i 指定源。

```
pip3 install protobuf==3.13.0
pip3 install psutil==5.7.0
pip3 install numpy
pip3 install scipy
pip3 install decorator==4.4.0
pip3 install sympy==1.5.1
pip3 install cffi==1.12.3
pip3 install pyyaml
pip3 install pathlib2 -i https://pypi.tuna.tsinghua.edu.cn/simple
```

依赖名称	版本号	安装命令
google.protobuf	>=3.13.0	pip3 install protobuf==3.13.0 --user
psutil	5.7.0	pip3 install psutil==5.7.0 --user
numpy	>=1.13.3	pip3 install numpy==1.13.3 --user
scipy	1.4.1	pip3 install scipy==1.4.1 --user
decorator	>=4.4.0	pip3 install decorator==4.4.0 --user
sympy	>=1.5.1	pip3 install sympy==1.5.1 --user
cffi	1.12.3	pip3 install cffi==1.12.3 --user
pyyaml	-	pip3 install pyyaml --user
pathlib2	-	pip3 install pathlib2 --user

(3) 安装 atc 软件

拷贝 SVP_PC/NNN_PC/Ascend-cann-toolkit_5.20.t6.2.b060_linux-x86_64.run 到虚拟机安装。更多参考官方 SDK 手册：ReleaseDoc\zh\01.software\pc\NNN\驱动和开发环境安装指南.pdf。

```
chmod +x Ascend-cann-toolkit_5.20.t6.2.b060_linux-x86_64.run
```

```
./Ascend-cann-toolkit_5.20.t6.2.b060_linux-x86_64.run --install
```

#{INSTALL_DIR}请替换为 CANN 软件安装后文件存储路径。例如，

\$HOME/Ascend/ascend-toolkit/latest/x86_64-linux。

```
source #{INSTALL_DIR}/bin/setenv.bash
```

```
sunshine@sunshine-vm: /workspace/Ascend$ chmod +x Ascend-cann-toolkit_5.20.t6.2.b060_linux-x86_64.run
sunshine@sunshine-vm: /workspace/Ascend$ ./Ascend-cann-toolkit_5.20.t6.2.b060_linux-x86_64.run --install
Verifying archive integrity... 100% SHA256 checksums are OK. 100% MD5 checksums are OK. All good.
Uncompressing 100%
[ToolKit] [20250902-10:40:04] [INFO] mkdir -p /home/sunshine/Ascend/ascend-toolkit/5.20.t6.2.b060/x86_64-linux
[ToolKit] [20250902-10:40:04] [INFO] mkdir -p /home/sunshine/Ascend/ascend-toolkit/5.20.t6.2.b060/arm64-linlinux200
[ToolKit] [20250902-10:40:04] [INFO] touch /home/sunshine/Ascend/ascend-toolkit/5.20.t6.2.b060/x86_64-linux/ascend_install.log
[ToolKit] [20250902-10:40:04] [INFO] install start
[ToolKit] [20250902-10:40:04] [INFO] The install path is /home/sunshine/Ascend !
[ToolKit] [20250902-10:40:04] [WARNING] driver package maybe not installed
[ToolKit] [20250902-10:40:04] [INFO] not find Huawei Device
[ToolKit] [20250902-10:40:04] [INFO] environment is not training or inference
[ToolKit] [20250902-10:40:04] [INFO] mkdir -p /home/sunshine/Ascend/ascend-toolkit/5.20.t6.2.b060/x86_64-linux/script
[ToolKit] [20250902-10:40:04] [INFO] install package CANN-runtime-5.20.t6.2.b060-linlinux200.run start
[ToolKit] [20250902-10:40:04] [INFO] ./run_package/CANN-runtime-5.20.t6.2.b060-linlinux200.run --devel --quiet --nox11 --install-path=/home/sunshine/Ascend/ascend-toolkit/5.20.t6.2.b060/arm64-linlinux200
[ToolKit] [20250902-10:40:09] [INFO] CANN-runtime-5.20.t6.2.b060-linlinux200.run --devel --quiet --nox11 install success
[ToolKit] [20250902-10:40:09] [INFO] install package CANN-compiler-5.20.t6.2.b060-ubuntu18.04.x86_64.run start
[ToolKit] [20250902-10:40:09] [INFO] ./run_package/CANN-compiler-5.20.t6.2.b060-ubuntu18.04.x86_64.run --devel --pylocal --quiet --nox11 --install-path=/home/sunshine/Ascend/ascend-toolkit/5.20.t6.2.b060/x86_64-linux
[ToolKit] [20250902-10:40:56] [INFO] CANN-compiler-5.20.t6.2.b060-ubuntu18.04.x86_64.run --devel --pylocal --quiet --nox11 install success
[ToolKit] [20250902-10:40:56] [INFO] install package CANN-runtime-5.20.t6.2.b060-ubuntu18.04.x86_64.run start
[ToolKit] [20250902-10:40:56] [INFO] ./run_package/CANN-runtime-5.20.t6.2.b060-ubuntu18.04.x86_64.run --devel --quiet --nox11 --install-path=/home/sunshine/Ascend/ascend-toolkit/5.20.t6.2.b060/x86_64-linux
[ToolKit] [20250902-10:40:59] [INFO] CANN-runtime-5.20.t6.2.b060-ubuntu18.04.x86_64.run --devel --quiet --nox11 install success
[ToolKit] [20250902-10:40:59] [INFO] install package CANN-app-5.20.t6.2.b060-ubuntu18.04.x86_64.run start
[ToolKit] [20250902-10:40:59] [INFO] ./run_package/CANN-app-5.20.t6.2.b060-ubuntu18.04.x86_64.run --devel --quiet --nox11 --install-path=/home/sunshine/Ascend/ascend-toolkit/5.20.t6.2.b060/x86_64-linux
[ToolKit] [20250902-10:41:24] [INFO] CANN-app-5.20.t6.2.b060-ubuntu18.04.x86_64.run --devel --quiet --nox11 install success
[ToolKit] [20250902-10:41:24] [INFO] install package CANN-toolkit-5.20.t6.2.b060-ubuntu18.04.x86_64.run start
[ToolKit] [20250902-10:41:24] [INFO] ./run_package/CANN-toolkit-5.20.t6.2.b060-ubuntu18.04.x86_64.run --devel --pylocal --quiet --nox11 --install-path=/home/sunshine/Ascend/ascend-toolkit/5.20.t6.2.b060/x86_64-linux
[ToolKit] [20250902-10:41:27] [INFO] CANN-toolkit-5.20.t6.2.b060-ubuntu18.04.x86_64.run --devel --pylocal --quiet --nox11 install success
[ToolKit] [20250902-10:41:27] [INFO] mkdir -p /home/sunshine/Ascend/ascend-toolkit/5.20.t6.2.b060/x86_64-linux/opp/op_tnpl/built-in/atcpu/atcpu_kernel/lib/LMIX
[ToolKit] [20250902-10:41:27] [INFO] copy ./lib/libcpu_kernels/context.a to /home/sunshine/Ascend/ascend-toolkit/5.20.t6.2.b060/x86_64-linux/opp/op_tnpl/built-in/atcpu/atcpu_kernel/lib/LMIX success.
[ToolKit] [20250902-10:41:27] [INFO] copy ./lib/libascend_protobuf.a to /home/sunshine/Ascend/ascend-toolkit/5.20.t6.2.b060/x86_64-linux/opp/op_tnpl/built-in/atcpu/atcpu_kernel/lib/LMIX success.
[ToolKit] [20250902-10:41:27] [INFO] create soft link success: /home/sunshine/Ascend/ascend-toolkit/5.20.t6.2.b060/runtime_linlinux200.aarch64 -> arm64-linlinux200/runtime
[ToolKit] [20250902-10:41:27] [INFO] create soft link success: /home/sunshine/Ascend/ascend-toolkit/5.20.t6.2.b060/acclib_linlinux200.aarch64 -> arm64-linlinux200/acclib
[ToolKit] [20250902-10:41:27] [INFO] create soft link success: /home/sunshine/Ascend/ascend-toolkit/5.20.t6.2.b060/acclib -> arm64-linlinux200/acclib
[ToolKit] [20250902-10:41:27] [INFO] create soft link success: /home/sunshine/Ascend/ascend-toolkit/5.20.t6.2.b060/compiler -> ./x86_64-linux/compiler
[ToolKit] [20250902-10:41:27] [INFO] create soft link success: /home/sunshine/Ascend/ascend-toolkit/5.20.t6.2.b060/atc -> ./x86_64-linux/atc
[ToolKit] [20250902-10:41:27] [INFO] create soft link success: /home/sunshine/Ascend/ascend-toolkit/5.20.t6.2.b060/opp -> ./x86_64-linux/opp
[ToolKit] [20250902-10:41:27] [INFO] create soft link success: /home/sunshine/Ascend/ascend-toolkit/5.20.t6.2.b060/toolkit -> ./x86_64-linux/toolkit
[ToolKit] [20250902-10:41:27] [INFO] create soft link success: /home/sunshine/Ascend/ascend-toolkit/latest -> ./5.20.t6.2.b060
[ToolKit] [20250902-10:41:27] [INFO] create soft link success: /home/sunshine/Ascend/ascend-toolkit/5.20 -> ./5.20.t6.2.b060
[ToolKit] [20250902-10:41:27] [INFO] create soft link success: /home/sunshine/Ascend/ascend-toolkit/5.20.t6.2.b060/x86_64-linux/ops -> ./opp
[ToolKit] [20250902-10:41:27] [INFO] Ascend-cann-toolkit_5.20.t6.2.b060_linux-x86_64.run install success,The install path is /home/sunshine/Ascend !
sunshine@sunshine-vm: /workspace/Ascend$
```

2.3. 模型转换

(1) 下载 yolov8 模型

[探索 Ultralytics YOLOv8 - Ultralytics YOLO 文档](#)

The screenshot displays the Ultralytics YOLOv8 website. On the left, a sidebar lists various models including YOLOv3, YOLOv4, YOLOv5, YOLOv6, YOLOv7, YOLOv8, YOLOv9, YOLOv10, YOLOv11, YOLOv12, YOLO12m, YOLO12n, YOLO12s, YOLO12xs, YOLO12xs-m, YOLO12xs-m2, YOLO12xs-m3, YOLO12xs-m4, YOLO12xs-m5, YOLO12xs-m6, YOLO12xs-m7, YOLO12xs-m8, YOLO12xs-m9, YOLO12xs-m10, YOLO12xs-m11, YOLO12xs-m12, YOLO12xs-m13, YOLO12xs-m14, YOLO12xs-m15, YOLO12xs-m16, YOLO12xs-m17, YOLO12xs-m18, YOLO12xs-m19, YOLO12xs-m20, YOLO12xs-m21, YOLO12xs-m22, YOLO12xs-m23, YOLO12xs-m24, YOLO12xs-m25, YOLO12xs-m26, YOLO12xs-m27, YOLO12xs-m28, YOLO12xs-m29, YOLO12xs-m30, YOLO12xs-m31, YOLO12xs-m32, YOLO12xs-m33, YOLO12xs-m34, YOLO12xs-m35, YOLO12xs-m36, YOLO12xs-m37, YOLO12xs-m38, YOLO12xs-m39, YOLO12xs-m40, YOLO12xs-m41, YOLO12xs-m42, YOLO12xs-m43, YOLO12xs-m44, YOLO12xs-m45, YOLO12xs-m46, YOLO12xs-m47, YOLO12xs-m48, YOLO12xs-m49, YOLO12xs-m50, YOLO12xs-m51, YOLO12xs-m52, YOLO12xs-m53, YOLO12xs-m54, YOLO12xs-m55, YOLO12xs-m56, YOLO12xs-m57, YOLO12xs-m58, YOLO12xs-m59, YOLO12xs-m60, YOLO12xs-m61, YOLO12xs-m62, YOLO12xs-m63, YOLO12xs-m64, YOLO12xs-m65, YOLO12xs-m66, YOLO12xs-m67, YOLO12xs-m68, YOLO12xs-m69, YOLO12xs-m70, YOLO12xs-m71, YOLO12xs-m72, YOLO12xs-m73, YOLO12xs-m74, YOLO12xs-m75, YOLO12xs-m76, YOLO12xs-m77, YOLO12xs-m78, YOLO12xs-m79, YOLO12xs-m80, YOLO12xs-m81, YOLO12xs-m82, YOLO12xs-m83, YOLO12xs-m84, YOLO12xs-m85, YOLO12xs-m86, YOLO12xs-m87, YOLO12xs-m88, YOLO12xs-m89, YOLO12xs-m90, YOLO12xs-m91, YOLO12xs-m92, YOLO12xs-m93, YOLO12xs-m94, YOLO12xs-m95, YOLO12xs-m96, YOLO12xs-m97, YOLO12xs-m98, YOLO12xs-m99, YOLO12xs-m100.

The main content area features a performance graph showing COCO AP@0.5 vs Latency T4 TensorRT FP16 (ms/img). Below the graph is a table of metrics for various models.

模型	尺寸 (像素)	mAP val 50-95	速度 CPU ONNX (秒张)	速度 A100 TensorRT (秒张)	参数 (M)	FLOPs (B)
YOLOv8n	640	37.3	80.4	0.99	3.2	8.7
YOLOv8s	640	44.9	128.4	1.20	11.2	28.6
YOLOv8m	640	50.2	234.7	1.83	25.9	78.9
YOLOv8l	640	52.9	375.2	2.39	43.7	165.2
YOLOv8x	640	53.9	479.1	3.53	68.2	257.8

Below the table, there is a section titled "YOLOv8 使用示例" (YOLOv8 Usage Examples) which provides a brief overview of the examples and links to the documentation.

(2) yolov8 模型测试(cnoda 进入 yolov8 环境)

a. 图片推理

```
from ultralytics import YOLO

import cv2

# 1. 加载模型

model = YOLO("yolov8n.pt")

# 2. 指定图片路径

img_path = "test.jpg"

# 3. 推理并保存结果

results = model(img_path)

# 4. 获取推理后的图片 (带标注)

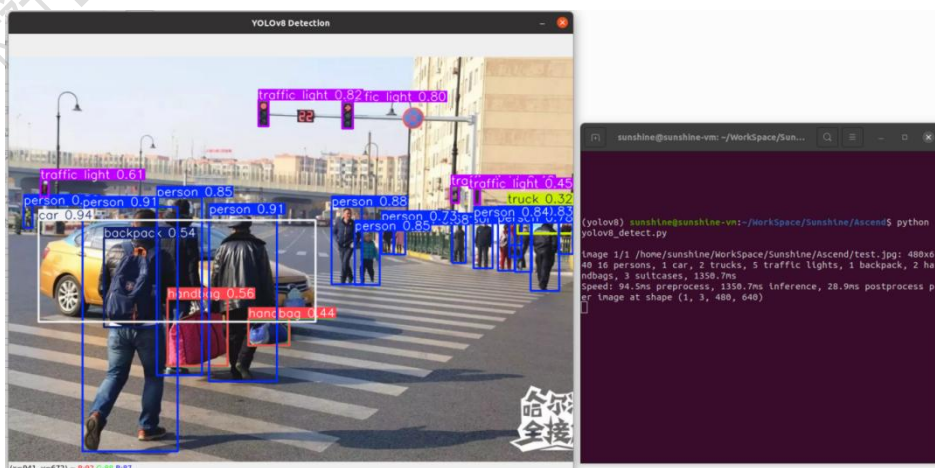
img_with_boxes = results[0].plot()

# 5. 显示窗口，阻塞等待用户按键

cv2.imshow("YOLOv8 Detection", img_with_boxes)

cv2.waitKey(0)

cv2.destroyAllWindows()
```



b. 打印所有类

```
from ultralytics import YOLO

# 1. 加载模型

model = YOLO("yolov8n.pt")

# 2. 获取类别字典

names = model.names # {id: name}

# 3. 打印所有类别

print("YOLOv8n.pt 模型包含的类别:")

for idx, name in names.items():

    print(f'{idx}: {name}')
```

```
(yolov8) sunshine@sunshine-vm:~/Workspace/Sunshine/Ascend$ python yolov8_classes.py
YOLOv8n.pt 模型包含的类别:
0: person
1: bicycle
2: car
3: motorcycle
4: airplane
5: bus
6: train
7: truck
8: boat
9: traffic light
10: fire hydrant
11: stop sign
12: parking meter
13: bench
14: bird
15: cat
16: dog
17: horse
18: sheep
19: dog
20: elephant
21: bear
22: zebra
23: giraffe
24: backpack
25: umbrella
26: handbag
27: tie
28: suitcase
29: frisbee
30: skis
31: snowboard
32: sports ball
33: kite
34: baseball bat
35: baseball glove
36: skateboard
37: surfboard
38: tennis racket
39: bottle
40: wine glass
41: cup
42: fork
43: knife
44: spoon
```

c. pt 转换 onnx

```
from ultralytics import YOLO

import cv2

model = YOLO("yolov8n.pt")

model.export(format="onnx", opset=12)
```

```
(yolov8) sunshine@sunshine-vm:~/Workspace/Sunshine/Ascend$ ls
bus.jpg  onnx2om.sh  op.cfg  pt2onnx.py  requirements_yolov8.txt  test.jpg  yolov8_classes.py  yolov8_detect.py  yolov8n.pt
(yolov8) sunshine@sunshine-vm:~/Workspace/Sunshine/Ascend$ python pt2onnx.py
Ultralytics 8.3.191 Python-3.10.18 torch-2.1.0+cu121 CPU (AMD Ryzen 5 2400G with Radeon Vega Graphics)
YOLOv8n summary (fused): 72 layers, 3,151,904 parameters, 0 gradients, 8.7 GFLOPs

PyTorch: starting from 'yolov8n.pt' with input shape (1, 3, 640, 640) BCHW and output shape(s) (1, 84, 8400) (6.2 MB)

ONNX: starting export with onnx 1.19.0 opset 12...
ONNX: slimming with onnxslim 0.1.65...
ONNX: export success 1.8s, saved as 'yolov8n.onnx' (12.2 MB)

Export complete (2.3s)
Results saved to /home/sunshine/Workspace/Sunshine/Ascend
Predict:      yolo predict task=detect model=yolov8n.onnx imgsz=640
Validate:     yolo val task=detect model=yolov8n.onnx imgsz=640 data=coco.yaml
Visualize:    https://netron.app
(yolov8) sunshine@sunshine-vm:~/Workspace/Sunshine/Ascend$ ls
bus.jpg  onnx2om.sh  op.cfg  pt2onnx.py  requirements_yolov8.txt  test.jpg  yolov8_classes.py  yolov8_detect.py  yolov8n.onnx  yolov8n.pt
(yolov8) sunshine@sunshine-vm:~/Workspace/Sunshine/Ascend$
```

(3) atc 工具使用(cnode 进入 atc 环境)

a. onnx 转换 om

```
#!/bin/sh

#source /home/sunshine/Ascend/ascend-toolkit/5.20/x86_64-linux/bin/setenv.bash

atc --model=$1 --framework=5 --output=$2 --soc_version="OPTG" --output_type=FP32

--insert_op_conf=./op.cfg
```

```
sunshine@sunshine-vm:~/Workspace/Sunshine/Ascend$ ls
bus.jpg  onnx2om.sh  op.cfg  pt2onnx.py  requirements_yolov8.txt  test.jpg  yolov8_classes.py  yolov8_detect.py  yolov8n.onnx  yolov8n.pt
sunshine@sunshine-vm:~/Workspace/Sunshine/Ascend$ conda activate atc
(atc) sunshine@sunshine-vm:~/Workspace/Sunshine/Ascend$ source /home/sunshine/Ascend/ascend-toolkit/5.20/x86_64-linux/bin/setenv.bash
(atc) sunshine@sunshine-vm:~/Workspace/Sunshine/Ascend$ ./onnx2om.sh yolov8n.onnx yolov8
ATC start working now, please wait for a moment.
ATC run success, welcome to the next use.

(atc) sunshine@sunshine-vm:~/Workspace/Sunshine/Ascend$ ls
bus.jpg  kernel_meta  op.cfg  requirements_yolov8.txt  yolov8_classes.py  yolov8n.onnx  yolov8n.om
fusion_result.json  onnx2om.sh  pt2onnx.py  test.jpg  yolov8_detect.py  yolov8n.pt
(atc) sunshine@sunshine-vm:~/Workspace/Sunshine/Ascend$
```

b. op.cfg

```
aipp_op {
  aipp_mode : static
  related_input_rank : 0
  max_src_image_size : 1228800
  support_rotation : false

  input_format : YUV420SP_U8 # RGB888_U8
  src_image_size_w : 640
  src_image_size_h : 640

  cpadding_value: 0.0
  csc_switch : true
  rbuv_swap_switch : false
  ax_swap_switch : false

  matrix_r0c0 : 256
  matrix_r0c1 : 0
  matrix_r0c2 : 359
  matrix_r1c0 : 256
  matrix_r1c1 : -88
  matrix_r1c2 : -183
  matrix_r2c0 : 256
  matrix_r2c1 : 454
  matrix_r2c2 : 0
  input_bias_0 : 0
  input_bias_1 : 128
  input_bias_2 : 128

  crop : false
  load_start_pos_h : 0
  load_start_pos_w : 0
  crop_size_w : 640
  crop_size_h : 640

  min_chn_0 : 0.0
  min_chn_1 : 0.0
  min_chn_2 : 0.0
  var_reci_chn_0 : 0.0039215686274509803921568627451
  var_reci_chn_1 : 0.0039215686274509803921568627451
  var_reci_chn_2 : 0.0039215686274509803921568627451
}
```

3. 开发板验证

测试环境为 V2.0.0 版本固件。

(1) 拷贝到板端并命名为 yolov8n.om， 或者修改代码。

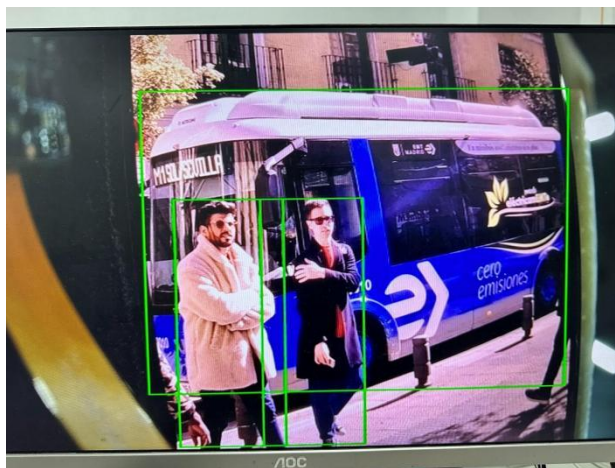
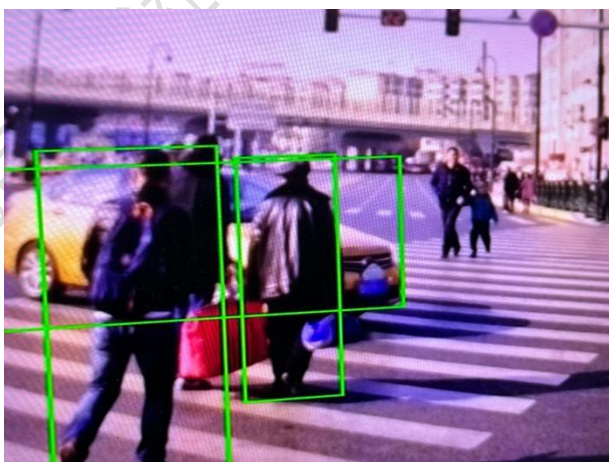
注：代码中默认只会在置信度 ≥ 0.85 时绘制框，可将阈值调低至 0.50 进行测试。

```
root@localhost:/opt/sample/yolov8# ls
sample_yolov8_imx347  sample_yolov8_os04a10  sample_yolov8_os08a20  yolov8.om
root@localhost:/opt/sample/yolov8# mv yolov8.om yolov8n.om
root@localhost:/opt/sample/yolov8# ls
sample_yolov8_imx347  sample_yolov8_os04a10  sample_yolov8_os08a20  yolov8n.om
root@localhost:/opt/sample/yolov8#
```

(2) 运行

```
export LD_LIBRARY_PATH=/opt/lib/npv:$LD_LIBRARY_PATH
./sample_yolov8_imx347
```

```
root@localhost:/opt/sample/yolov8# export LD_LIBRARY_PATH=/opt/lib/npv:$LD_LIBRARY_PATH
root@localhost:/opt/sample/yolov8# ./sample_yolov8_imx347
[EVENT] CCECPU(2157,sample_yolov8_imx347):2025-09-03-06:20:20.506.327 [aicpusd_cust_so_manager.cpp:73][AicpuCustSoManagerInit][tid:2157] cust so dir name is /root/cust_aicpu_0_2157/.
linear mode
IMX347 Slave 4M 30fps 12bit LINE Init OK!===
ISP Dev 0 running !
-----press Enter key to exit!-----
[EVENT] PROFILING(2157,sample_yolov8_imx347):2025-09-03-06:20:25.026.834 [msprof_callback_impl.cpp:99] >>> (tid:2164) MsprofCtrlCallback called, type: 255
[EVENT] PROFILING(2157,sample_yolov8_imx347):2025-09-03-06:20:25.027.226 [prof_acl_mgr.cpp:1333] >>> (tid:2164) Init profiling for dynamic profiling
[level]:Info,[func]:sample_svp_npu_init_resource [line]:92 [info]:acl init success.
[EVENT] CCECPU(2157,sample_yolov8_imx347):2025-09-03-06:20:25.193.367 [ae_so_manager.cc:440][LoadSo][tid:2164][AICPU_PROCESSER] Single so manager init failed, soFile is /usr/lib64/aicpu_kernels/0/aicpu_kernels_device/liba1cpu_kernels.so
[ERROR] CCECPU(2157,sample_yolov8_imx347):2025-09-03-06:20:25.193.571 [ae_so_manager.cc:453][LoadSo][tid:2164][AICPU_PROCESSER] Single so manager init failed, soFile is /usr/lib64/aicpu_kernels/liba1cpu_kernels.so
[EVENT] CCECPU(2157,sample_yolov8_imx347):2025-09-03-06:20:25.193.632 [ae_kernel_lib_aicpu.cc:168][BatchLoadKernelSo][tid:2164][AICPU_PROCESSER] Load so liba1cpu_kernels.so failed.
[EVENT] CCECPU(2157,sample_yolov8_imx347):2025-09-03-06:20:25.193.694 [aicpusd_interface.cpp:114][AICPUPreOpenKernels][tid:2164] Preload so liba1cpu_kernels.so success.
[EVENT] CCECPU(2157,sample_yolov8_imx347):2025-09-03-06:20:25.193.776 [ae_so_manager.cc:440][LoadSo][tid:2164][AICPU_PROCESSER] Single so manager init failed, soFile is /usr/lib64/aicpu_kernels/0/aicpu_kernels_device/libcpu_kernels.so
[ERROR] CCECPU(2157,sample_yolov8_imx347):2025-09-03-06:20:25.193.845 [ae_so_manager.cc:453][LoadSo][tid:2164][AICPU_PROCESSER] Single so manager init failed, soFile is /usr/lib64/aicpu_kernels/libcpu_kernels.so
[EVENT] CCECPU(2157,sample_yolov8_imx347):2025-09-03-06:20:25.193.903 [ae_kernel_lib_aicpu.cc:168][BatchLoadKernelSo][tid:2164][AICPU_PROCESSER] Load so libcpu_kernels.so failed.
[EVENT] CCECPU(2157,sample_yolov8_imx347):2025-09-03-06:20:25.193.959 [aicpusd_interface.cpp:114][AICPUPreOpenKernels][tid:2164] Preload so libcpu_kernels.so success.
[EVENT] PROFILING(2157,sample_yolov8_imx347):2025-09-03-06:20:25.204.725 [msprof_callback_impl.cpp:120] >>> (tid:2164) MsprofSetDeviceCallback called, is open: 1
[level]:Info,[func]:sample_svp_npu_init_resource [line]:99 [info]:open device 0 success.
[level]:Info,[func]:sample_svp_npu_init_resource [line]:186 [info]:get run mode success
[level]:Info,[func]:sample_npu_load_model_with_mem [line]:165 [info]:load mem size:63120896 weight size:51859968 id:1
[level]:Info,[func]:sample_npu_load_model_with_mem [line]:168 [info]:load model /opt/sample/yolov8/yolov8n.om success
[level]:Info,[func]:sample_npu_create_desc [line]:143 [info]:create model description success
sample_svp_npu_load_model_0
get frame success: w:h, 640x640
[level]:Info,[func]:sample_npu_create_input_dataset [line]:184 [info]:create model input dataset success
[level]:Info,[func]:sample_npu_create_output [line]:421 [info]:create model output to SUCCESS
[level]:Info,[func]:sample_npu_create_input_databuf [line]:227 [info]:create model input success
[level]:Info,[func]:sample_npu_model_execute [line]:541 [info]:end aclmdlExecute, modelId is 1
75 = 0.73975 at 531.00 2.00 108.00 x 627.00 vase 74.0e
[level]:Info,[func]:sample_npu_output_model_result_yolo [line]:515 [info]:output data success
[level]:Info,[func]:sample_npu_destroy_input_databuf [line]:245 [info]:destroy model input data buf success
[level]:Info,[func]:sample_npu_destroy_input_dataset [line]:197 [info]:destroy model input dataset success
75 = 0.73975 at 531.00 2.00 108.00
```





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